5.7 Inverse of a Function

Name _____ Period ____ Date _____

Algebra II 5.7 (30-minute plan) · Student sheet

I can…

- find the inverse of a linear function.
- find the inverse of a nonlinear function, restricting the domain when needed.
- verify that two functions are inverses.

Vocabulary · fill in the missing word

Inverse functions — functions that _____ each other; f^{-1} reverses the inputs and outputs of f

Finding an inverse — _____ x and y, then solve for y

Horizontal line test — the inverse of f is a _____ when no horizontal line crosses the graph of f more than once

Verifying inverses — f(g(x)) = _____ = x

Part 1 · Notes and examples (fill in each note, then work the example)

Example 1

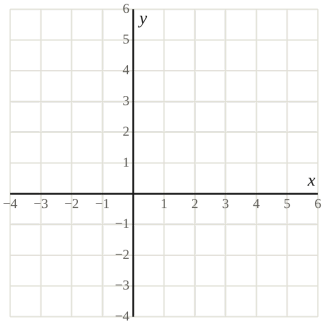
Let f(x) = 2x + 3. Solve y = f(x) for x. Then find the input when the output is −7.

Finding an inverse: To find an inverse: replace f(x) with y, _____ x and y, then solve for _____.

Graphs of inverses: The graphs of f and f^{-1} are reflections in the line _____.

Example 2

Find the inverse of f(x) = 3x − 1. Then graph f and f^{-1}.

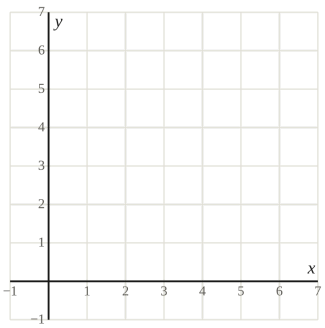


Horizontal line test: the inverse of f is a function when no horizontal line crosses the graph of f more than _____.

Verifying inverses: To verify that f and g are inverses, show that _____ = x and _____ = x.

Example 3

Find the inverse of $f(x) = x^2 + 2$, $x \geq 0$. Why is the restriction needed?



Part 2 · Partner practice

Partner practice

Find the inverse of $f(x) = -4x + 8$.